

XRPL Validator — Daily Engineering Report • 2026-07-19

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **8 commits** (xrpl-validator 8)

Engineering changes

- **feat(xrpl-ledger): order-book quality keylet + OfferCreate placement — base camp** (150c432 , xrpl-validator) The book quality directory key was the last missing keylet family: - book_base = SHA512Half(0x0042 'B' + pays_cur + gets_cur + pays_iss + gets_iss); quality dir = base[0..24] + quality_be64 - offer_quality mirrors rippled getRate under the STNumber switchover: normalize both mantissas to [1e15,1e16), truncating muldiv at 10^{17} , +5, then ONE round-half-even pass over the excess tail (Number canonicalize). Locked ...
- **feat(xrpl-ledger): integer offer-crossing engine — the summit assault** (7ee6dcd , xrpl-validator) Replaces the f64 crossing stub with a real integer engine: - SHAMap::keys_with_prefix (nibble-radix descent) + Sandbox merged view — native book enumeration: walk a book's quality dirs ascending from best quality, stop past the taker's threshold (swapped-args offer_quality) - (mantissa u128, exponent i32) arithmetic with directed rounding at fill boundaries (taker pays UP, receives DOWN — Quality::ceil_in/out ...
- **fix(xrpl-ledger): cancel-and-replace + funded-check starvation — OfferCreate 64%** (f5b87b2 , xrpl-validator) Two data-diagnosed fixes to the crossing engine's first corpus run: - OfferSequence on OfferCreate cancels the named prior offer BEFORE crossing (rippled cancel-and-replace; the '126 crossing misses' were the taker's own old offers — replaced, not crossed) - differential read-set loads the taker's gets-side trust line: a pure placement never has it in mainnet meta, so available() starved and threw 220 false ...
- **feat(xrpl-ledger): IOU + cross-currency payment engine — second summit** (e6efe32 , xrpl-validator) Payment was XRP-only (preflight returned tecUNSUPPORTED for IOU; preclaim demanded an XRP balance even for cross-currency payments BUYING XRP). Segmentation showed the corpus is dominated by arb-probe partial payments (XRP-deliver, IOU SendMax, tfPartialPayment) that mainnet fails tecPATH_DRY fee-only — natives just needed to attempt the path and agree.
- **feat(xrpl-ledger): CheckCash via shared transfer engine — IOU checks + tecPATH_PARTIAL semantics** (a4dc8d7 , xrpl-validator) CheckCash now routes through the offer.rs transfer helpers (leg_of/available/move_leg) instead of hand-rolled XRP-drops balance edits: IOU checks cash over trust lines, issuer self-cover is unlimited, and writer shortfall fails the rippled way (tecPATH_PARTIAL, fee-only) instead of tecUNFUNDED. Check deletion now unlinks BOTH owner directories (OwnerNode + DestinationNode hints) and decrements both owner counts. ...
- **feat(xrpl-ledger): CheckCash via transfer engine + tec classification — corpus 90%** (02508d9 , xrpl-validator)

- CheckCash rides the shared engine: SendMax fixes the currency, writer shortfall (or over-asking) fails the rippled way (tecPATH_PARTIAL, fee-only), success moves the leg + deletes the check + unlinks BOTH owner directories (OwnerNode/DestinationNode hints) + OwnerCounts. CheckCash 0/30 -> 30/30. - probe read-set loads the CheckID object (a fee-only mainnet meta never contains it -> native said tecNO_ENTRY where ...
- **fix(xrpl-ledger): path payments — quality gate only under tfLimitQuality, DeliverMin floor, book pre-state — corpus 90.3%** (28f0cb6 , xrpl-validator) Two root causes behind the 62 tecPATH_DRY divergences: - Engine: apply_path_payment imposed the SendMax/Amount ratio as a per-offer quality bound unconditionally. rippled only does that under tfLimitQuality; arb-style conversions use a sentinel-max Amount whose implied ratio reads every book as dry. Threshold is now unlimited without the flag (aggregate SendMax/Amount bounds still hold), and DeliverMin is enforced ...
- **feat(xrpl-ledger): AMM swap engine — corpus 94.1%, the third summit** (18d4a00 , xrpl-validator) New amm_swap module: faithful port of rippled AMMHelpers under fixAMMv1_1/v1_2. A 16-significant-digit decimal Number core with per-op directed rounding (correctly rounded mul/div/add/sub/sqrt), swapAssetIn/swapAssetOut with the amendment's AMM-favoring rounding sequence, spot- quality gating vs the book (strictly-better + 1e-7 relative threshold), book-anchored sizing via changeSpotPriceQuality (citardauq ...

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Selective-Disclosure Merkle Audit Trails** (audit) Add a new owner-reserved ledger object `ltAUDIT_COMMITMENT` (keylet::auditCommitment(owner, sequence)) holding an append-only Merkle Mountain Range ro
- **Time-Limited NFT Rental Escrow** (nft) Add a `tfRental` flag and `sfRentExpiration` (LedgerIndex) SField to `NFTTokenEscrowCreate` (XLS-51). When set, `EscrowFinish` does not reassign the `N
- **Whitelisted On-Ledger Parameter Voting** (governance) Add a singleton-per-key, reserve-free ledger object `ltPARAMETER` (keylet::parameter(name), same footing as `Amendments / FeeSettings`) storing `sfPar

Appendix — commits

Time	Repo	Commit	Subject
07:40	xrpl-validator	150c432	feat(xrpl-ledger): order-book quality keylet + OfferCreate placement — base camp
08:38	xrpl-validator	7ee6dcd	feat(xrpl-ledger): integer offer-crossing engine — the summit assault
09:34	xrpl-validator	f5b87b2	fix(xrpl-ledger): cancel-and-replace + funded-check starvation — OfferCreate 64%
10:43	xrpl-validator	e6efe32	feat(xrpl-ledger): IOU + cross-currency payment engine — second summit
11:58	xrpl-validator	a4dc8d7	feat(xrpl-ledger): CheckCash via shared transfer engine — IOU checks + tecPATH_PARTIAL semantics

Time	Repo	Commit	Subject
12:29	xrpl-validator	02508d9	feat(xrpl-ledger): CheckCash via transfer engine + tec classification — corpus 90%
15:50	xrpl-validator	28f0cb6	fix(xrpl-ledger): path payments — quality gate only under tfLimitQuality, DeliverMin floor, book pre-state — corpus 90.3%
20:12	xrpl-validator	18d4a00	feat(xrpl-ledger): AMM swap engine — corpus 94.1%, the third summit